

Three-Origin Triangulation of a Finite Prime–Shell Spectral Ladder

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Abstract

Finite spectral experiments can be tied to one residue environment even when their summary looks stable. We therefore add a third, disjoint source origin to the frozen TPC–325/TPC–326 four-rung ladder and record a three-origin envelope range. The literal deleted-diagonal centered prime-shell blocks, height, shell anchors, exponents, and four sign laws are unchanged. At the new origin 20001, all-plus normalized profile majorization holds on all 32 rows, and both the profile and energy censuses agree with the two earlier origins. The maximum pooled TV and energy ranges are 0.0007970083 and 0.0045518412, below the predeclared controls 0.001 and 0.005. Reverse/einsum reconstruction, residue perturbation stress, exact rational anchoring, and normal/optimized replay are included. The result is finite triangulation evidence only: it does not prove a source-uniform limit, arithmetic L^2 cancellation, power saving, or twin-prime conclusion.

1 Question and frozen object

TPC–325 and TPC–326 used the same finite prime-shell operator at origins 12001 and 16001. The next minimal adversarial question is whether their finite readout survives a third residue environment and whether the agreement can be summarized without choosing one panel as the reference. We use

$$I_N = [20001, 20000 + N/2] \cap \mathbb{Z}, \quad N \in \{320, 640, 1280, 2560\}.$$

For $p \in (Q, 2Q]$, define

$$B_{p,N}^{(s)}(u, t) = p \frac{66^{2s}}{(66^2 + (u-t)^2)^s} \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right) \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t}. \quad (1)$$

The direct and signed Grams are

$$G_0(N) = \sum_p B_p^* B_p, \quad G_e(N) = \left(\sum_p e_p B_p \right)^* \left(\sum_p e_p B_p \right).$$

The sign menu is all-plus, index-alternating, the mod-4 character, and a half split. The shell anchors are $Q \in \{24, 36, 54, 80\}$ and $s \in \{1, 2\}$.

For a positive-trace Gram G , let

$$\pi(G) = \left(\frac{\lambda_1(G)}{\text{tr } G}, \dots, \frac{\lambda_d(G)}{\text{tr } G} \right), \quad \lambda_1 \geq \dots \geq \lambda_d.$$

Finite Gram positivity and trace normalization give the basic spectral typing used here; see [1]. No asymptotic statement is hidden in this definition.

2 Triangulation protocol

The four new intervals are

$$[20001, 20160] \subset [20001, 20320] \subset [20001, 20640] \subset [20001, 21280].$$

They are disjoint from the earlier ladders and from the older source panels. The only intervention relative to the locked parents is the origin. The three-origin diagnostic at scale N is

$$\text{range}_T(N) = \max_{o \in \{12001, 16001, 20001\}} T_o(N) - \min_o T_o(N),$$

where T_o is the all-plus TV lower envelope over the eight (Q, s) rows. The energy range $\text{range}_E(N)$ is defined analogously using the all-plus energy upper envelope. We require the maximum ranges to be strictly below 0.001 and 0.005, respectively, and require them to be nonzero as a non-vacuity control.

3 Finite results

Table 1: New-origin all-plus ladder.

N	$ I_N $	TV lower envelope	energy upper envelope
320	160	0.2852340552	8.901456172
640	320	0.2108709647	6.864102783
1280	640	0.1900525186	6.249238483
2560	1280	0.1700854483	5.998451633

All 32 new rows have positive lower endpoints in the interior prefix test, and all-plus profile majorization holds on 32/32. The four-law profile census, listed as ‘(signed majorizes, mixed)’, is

law	signed majorizes	mixed
all-plus	32	0
alternating index	21	11
mod-4 character	26	6
half split	23	9

with zero direct-majorization and unresolved entries. These counts agree with both earlier certificates. The energy-side counts also agree: all-plus is below one on 4/32 rows and above one on 28/32; the other laws retain their parent counts.

Table 2: Three-origin pooled envelope ranges.

N	$\text{range}_T(N)$	$\text{range}_E(N)$
320	0.0007970083	0.0045518412
640	0.0003338713	0.0044707079
1280	0.0001938660	0.0013977240
2560	0.0000914121	0.0006212834

The maximum ranges are below the frozen controls, while every displayed range is positive. Thus the result is a finite three-point triangulation, not a copied equality. At the exact rational anchor [20001, 20016] with $Q = 4$ and $s = 1$, the shell is $\{5, 7\}$. The direct and alternating signed energy digests are respectively

97225bdbd0cb628956b3701748cec3b2eca7b4d559c0d0b42044300f7c26889b,

f38ac7229026dcd2ada592c5b245871d3ef1856e4bac21c86010e89766a9f9f7.

The certificate retains the exact rational numerator/denominator identities; the independent checker recomputes them without importing the producer.

4 Interpretation and limitations

The strongest justified statement is

NUMERICALLY_CERTIFIED_FINITE_THREE_ORIGIN_SCALE_TRIANGULATION.

The third origin does not falsify the earlier finite profile readout, and the pooled ranges quantify its finite source sensitivity. It does not establish uniformity over source origins or over a growing scale ladder. In particular, the operator has no source-native Möbius or von Mangoldt signed estimate, so

ARITHMETIC_ADVANCE=NO, FIXED_POWER_CREDIT=0, FULL_GATE_B=OPEN.

The Session-named Route-A and Route-B evaluators are absent from this checkout; the local Bridge-B checker is therefore a fail-closed fallback, not an official evaluator pass. No twin-prime conclusion is asserted.

5 Reproducibility

The project contains the locked parent artifacts, canonical full-row certificate, independent reverse/einsum checker, residue stress suite, proof and theorem ledgers, and this manuscript. From the repository root run:

```
python -B papers/tpc-327-three-origin-scale-triangulation/code/
    tpc327_three_origin_scale_triangulation.py --check
python -B papers/tpc-327-three-origin-scale-triangulation/experiments/
    tpc327_independent_checker.py --check
python -B papers/tpc-327-three-origin-scale-triangulation/experiments/
    tpc327_three_origin_stress.py --check
```

Normal and optimized runs, PDF hygiene, and stdout equality are enforced by the local Bridge-B checker.

References

- [1] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2013.